

Aussie Primary Academy | Australian Curriculum v9.0 | Learn, Investigate and Explain

Name: _____

Date: _____

LEARNING INTENTION	identify sources of heat and investigate how heat energy transfers from warmer objects or places to cooler ones
KEY IDEA	Heat energy moves from warmer to cooler objects or regions until temperatures become more similar. Sources include the sun, electricity, burning and friction.
WORKED MODEL	A metal spoon in warm soup becomes warmer because heat transfers from the hotter soup into the cooler spoon.

SUCCESS CRITERIA - I CAN

[] use key vocabulary [] classify or model accurately [] explain using evidence [] correct a misconception

A. BUILD THE SCIENCE IDEA

1. Define and use these terms: heat source and temperature.

2. Match examples to: heat source, temperature, transfer.

3. Annotate the worked model with two observations and one scientific explanation.

B. APPLY AND EXPLAIN

4. Sort six examples from this context: sun-warmed pavement, a toaster, rubbing hands, ice in water, a metal spoon and a wool wrap

5. Explain two classifications using observable evidence, not appearance alone.

6. Identify the changed condition, the effect and one condition that should stay the same.

C. INDEPENDENT CHECK

7. Draw or construct a labelled scientific model that shows the relationship.

8. Write a three-sentence explanation using because, therefore and evidence.

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D. APPLY SCIENCE IN CONTEXT

9. Case study: Three cups of warm water are wrapped in foil, wool or paper and measured every five minutes.

10. Record the important observations in a two-column evidence table.

11. Write a claim-evidence-reasoning conclusion and state one limitation.

E. REASON, EVALUATE AND TRANSFER

12. Find and correct the inaccurate statement in the case study; explain why it is inaccurate.

13. Apply the same science idea to a different everyday situation and predict what would happen.

F. STUDENT REVIEW - SHOW WHAT YOU UNDERSTAND

[] I can identify sources of heat and investigate how heat energy transfers from warmer objects or places to cooler ones.

Circle one: independently | with one reminder | with support

Evidence from my work that proves I understand: _____

The most difficult science idea was _____ because _____

One correction or improvement I made: _____

A question I still have: _____

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Accept equivalent responses when the student uses accurate science ideas, relevant evidence and clear reasoning.

COMPLETE ANSWER KEY

1. Source produces heat; temperature measures hotness; transfer is energy movement.
2. Sun/toaster/friction are sources; remaining examples show transfer or insulation.
3. Heat moves from warmer soup to cooler spoon.
4. Sort by source, transfer or reduced transfer.
5. Reasons must name temperature difference and direction.
6. Changed: wrap; measured: temperature change; controls include cup, volume, start temperature and time.
- 7-8. Arrows point warmer to cooler; explanation uses evidence.
9. Compare temperature drops to identify the best insulator.
10. Table should show material and change over time.
11. Small trials and uncontrolled room conditions limit certainty.
12. Heat does not rise in every situation; it transfers in several ways from warmer to cooler.
13. Accept a reasoned prediction for clothing, cooking or buildings.

TEACHER CHECK AND NEXT STEP

[] Uses accurate vocabulary and a complete cause-and-effect explanation.

[] Supports conclusions with observations or data.

Next step: revisit one misconception, then transfer the idea to a new example.

COMMON MISCONCEPTION / WATCH OUT

Cold does not flow into an object; heat transfers out of the warmer object.

Temperature and heat are related but are not interchangeable words.

EXTENSION AND RE-TEACH

Re-teach: model one difficult item, complete one together, then ask the student to explain an independent correction.

Extension: apply the science idea to a new real-world example and justify the explanation with observations or data.